Utilization of a health-system specialty pharmacy to optimize management of HCV-positive kidney transplant recipients

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Background

- Over 100,000 patients in the U.S. are awaiting a kidney transplant, and the average wait time is approximately 3 to 5 years.[1]
- Historically, individuals living with hepatitis C virus (HCV) have been underutilized as donors due to concerns related to disease transmission.[2]
- Advances in HCV treatment have led many centers to consider these individuals as donors to help reduce transplant wait times.[2]
- Health-system specialty pharmacy (HSSP) teams can play an important role in the kidney transplant process to improve medication access by addressing cost barriers for direct-acting antivirals (DAAs) and minimizing delays in therapy initiation.

Objective

To describe the development of an HSSP program for the management of kidney transplant recipients receiving a kidney allograft from a donor with a positive HCV nucleic acid test (HCV-NAT+).

Description

Quality Improvement Initiative

This quality improvement initiative was conducted at a health system located in Portland, Oregon from November 2024 to May 2025.

Evaluation Measures

- Number of patients who elected to receive an HCV-NAT+ kidney allograft and were referred to the HSSP for DAA initiation
- Prior authorization approval rate
- Number of patients receiving financial assistance
- Mean time to treatment initiation

Data Analysis

- Descriptive statistics were used to present all measures.

Description

Figure 1: Workflow for Patients Receiving an HCV-NAT+ Kidney Allograft

Pre-transplant

1. Patient elects to receive an HCV-NAT+ kidney allograft
2. Multidisciplinary transplant team evaluates treatment options and chooses a DAA
3. Transplant team refers the patient to the HSSP
4. HSSP team completes a benefits investigation
5. HSSP team completes any required prior authorization
6. HSSP team secures financial assistance for patients in need

Post-transplant

1. Medication delivery or pick-up is coordinated with the patient
2. HSSP pharmacist provides initial counseling to ensure the patient understands how to safely and effectively use the medication
3. HSSP checks in with patient at each medication fill and discusses any medication-related issues such as adherence or side effects
4. HSSP team monitors safety and effectiveness of therapy based on standardized clinical protocols
5. If medication issues arise during treatment, a pharmacist provides support by: (1) offering additional education, (2) recommending a therapy change to the prescriber, or (3) referring the patient to see the prescriber

Evaluation

To date, three patients who elected to receive an HCV-NAT+ kidney allograft have been referred to the HSSP for DAA initiation. The HSSP team successfully secured financial assistance for two patients.

100% Prior authorization approval rate

7 days Mean time to treatment initiation

Discussion & Conclusion

- Given the demand for kidney transplants far exceeds the available organ supply, utilizing HCV-NAT+ donors can help expand the donor pool and shorten wait times for recipients.
- HSSP teams have expertise in supporting medication access and managing complex conditions.
- HSSP teams can be valuable assets in the clinical management of HCV-NAT+ kidney transplant recipients.

Next Steps

- Future studies should evaluate clinical outcomes for patients receiving HCV-NAT+ kidney allografts who initiate DAAs, including the impact of treatment delays.

References

1. Organ Procurement and Transplantation Network (OPTN) and Scientific Registry of Transplant Recipients (SRTR) OPTN/SRTR 2013 annual data report. Rockville, MD: Department of Health and Human Services, Health Resources and Services Administration. *Am J Transplant*. 2015 Jan;15(S2):4–15.

2. Patnaik R, Tsai E. Hepatitis C virus treatment and solid organ transplantation. *Gastroenterol Hepatol (NY)*. 2022;18(2):85-94. PMID: 35505819